

MUTUAL FUND ANALYTICS

Capstone Project I

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Submitted To:

Bluestock Fintech

Project Duration: 20 June 2026 – 10 July 2026

Technologies Used:

- Python
- Pandas
- NumPy
- SQLite
- SQL
- Matplotlib
- Power BI
- GitHub

Submission Date: 10 July 2026

Executive Summary

The Mutual Fund Analytics project is an end-to-end data analytics solution developed to analyse the performance of Indian mutual funds, investor behaviour, portfolio allocation, and market trends using Python, SQL, SQLite, and Power BI. The project aims to

transform raw financial datasets into meaningful insights that can support informed investment decisions.

The project begins with the collection and preprocessing of multiple mutual fund datasets, including fund master data, NAV history, Assets Under Management (AUM), SIP inflows, scheme performance, investor transactions, portfolio holdings, and benchmark indices. An automated ETL pipeline was developed to clean, validate, and transform the raw datasets before storing them in a structured SQLite database.

Exploratory Data Analysis (EDA) was performed to identify trends in mutual fund performance, investor demographics, transaction patterns, category-wise fund growth, and portfolio allocation. SQL queries were used to extract meaningful business insights and support analytical reporting.

The project also includes comprehensive performance analytics by calculating financial metrics such as Annual Returns, Alpha, Beta, Sharpe Ratio, Sortino Ratio, Standard Deviation, Maximum Drawdown, Value at Risk (VaR), and Conditional Value at Risk (CVaR). Advanced analytical techniques including Investor Cohort Analysis, SIP Continuity Analysis, Portfolio Diversification Analysis, and a Mutual Fund Recommendation System were implemented to evaluate investor behaviour and fund performance.

An interactive four-page Power BI dashboard was developed to visualize key performance indicators, fund performance, investor analytics, SIP trends, and market insights. The dashboard enables users to explore the data through interactive filters and visualizations, making complex financial information easier to understand.

Overall, this project demonstrates the practical application of Data Engineering, Data Analysis, Financial Analytics, SQL, Business Intelligence, and Data Visualization to solve real-world problems in the mutual fund industry. The solution provides investors and analysts with a comprehensive framework for evaluating mutual fund performance and making data-driven investment decisions.

Project Objectives

The primary objective of this project is to develop an end-to-end Mutual Fund Analytics solution capable of transforming raw mutual fund data into meaningful insights using data engineering, financial analytics, and business intelligence techniques. The project focuses on analysing fund performance, investor behaviour, portfolio allocation, and market trends to support better investment decision-making.

Objectives

- Develop an automated ETL pipeline to collect, clean, and preprocess mutual fund datasets.
- Store cleaned data in a structured SQLite relational database for efficient querying.
- Perform SQL-based analytical queries to extract business insights.
- Conduct Exploratory Data Analysis (EDA) to identify trends and patterns in mutual fund performance.
- Calculate important financial performance metrics such as Annual Return, Sharpe Ratio, Alpha, Beta, Sortino Ratio, Maximum Drawdown, Value at Risk (VaR), and Conditional Value at Risk (CVaR).
- Analyse investor behaviour using Investor Cohort Analysis and SIP Continuity Analysis.
- Evaluate portfolio diversification and sector allocation.
- Develop a basic recommendation system to suggest mutual funds based on quantitative performance metrics.
- Design an interactive four-page Power BI dashboard for visualizing fund performance, investor analytics, SIP trends, and market insights.
- Generate business recommendations that assist investors in making informed investment decisions.

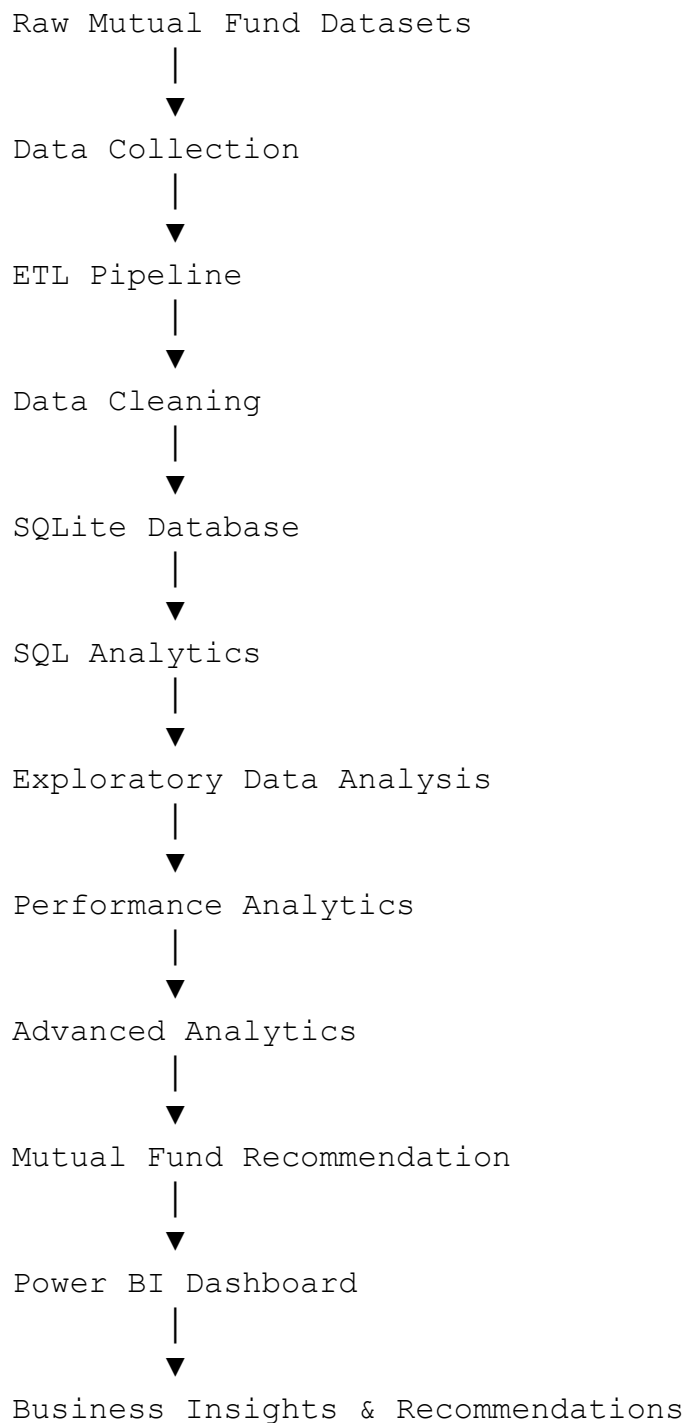
Project Overview

The Mutual Fund Analytics project is an end-to-end financial data analytics solution developed to analyse Indian mutual fund performance, investor behaviour, portfolio allocation, and market trends. The project integrates data engineering, database management, financial analytics, and business intelligence techniques to transform raw mutual fund data into meaningful insights that support investment decision-making.

The project starts with collecting multiple mutual fund datasets including Fund Master, NAV History, AUM, SIP inflows, investor transactions, portfolio holdings, scheme performance, and benchmark indices. An automated ETL pipeline cleans, validates, and transforms the data before storing it in a structured SQLite database. SQL queries and Exploratory Data Analysis (EDA) are then used to identify market trends, fund performance, and investor behaviour. Advanced analytics such as Sharpe Ratio, Alpha, Beta, Sortino Ratio, Maximum Drawdown, Value at Risk (VaR), Conditional Value at Risk

(CVaR), Investor Cohort Analysis, SIP Continuity Analysis, Portfolio Diversification Analysis, and a Mutual Fund Recommendation System are implemented to evaluate investment performance and risk. Finally, an interactive four-page Power BI dashboard presents the results through dynamic visualizations and business insights.

Project Workflow



Technologies Used

Technology	Purpose
Python	Data preprocessing, analytics and automation
Pandas	Data cleaning and manipulation
NumPy	Numerical computations
SQLite	Relational database management
SQL	Analytical querying
Matplotlib	Data visualization
Power BI	Interactive dashboard development
Git & GitHub	Version control
Jupyter Notebook	Analysis and documentation

Figure 1: Project Directory Structure

Name	Status	Date modified	Type	Size
dashboard	✓	02-07-2026 01:49	File folder	
data	✓	02-07-2026 00:53	File folder	
notebooks	✓	02-07-2026 00:55	File folder	
reports	✓	02-07-2026 00:04	File folder	
scripts	✓	02-07-2026 01:10	File folder	
sql	✓	24-06-2026 15:43	File folder	
.gitignore	✓	02-07-2026 01:49	Git Ignore Source ...	1 KB
bluestock_mf	✓	02-07-2026 01:37	Data Base File	4,676 KB
README	✓	02-07-2026 01:39	Markdown Source ...	5 KB
requirements	✓	22-06-2026 01:14	Text Document	1 KB

Dataset Description

Data Sources

The Mutual Fund Analytics project utilizes multiple datasets covering different aspects of the Indian mutual fund industry. These datasets include fund master information, historical NAV prices, Assets Under Management (AUM), SIP inflows, category-wise investments, investor transactions, portfolio holdings, benchmark indices, and live NAV data collected through MFAPI. These datasets provide a comprehensive view of fund performance, investor behaviour, and market trends.

Dataset Summary

Dataset	Description
01_fund_master.csv	Mutual fund master details
02_nav_history.csv	Historical NAV records

03_aum_by_fund_house.csv	Assets Under Management by AMC
04_monthly_sip_inflows.csv	Monthly SIP inflow statistics
05_category_inflows.csv	Category-wise investment inflows
06_industry_folio_count.csv	Industry folio statistics
07_scheme_performance.csv	Scheme returns and risk metrics
08_investor_transactions.csv	Investor transaction records
09_portfolio_holdings.csv	Portfolio holdings and sector allocation
10_benchmark_indices.csv	Benchmark index performance
Live NAV Files	Real-time NAV fetched using MFAPI

Data Cleaning and Preprocessing

Before analysis, all datasets underwent preprocessing to ensure consistency and accuracy. The cleaning process included handling missing values, removing duplicate records, correcting data types, converting date columns into datetime format, validating numerical fields, standardizing categorical values, and preparing processed datasets for analysis. The cleaned files were stored in the 'data/processed' directory.

Database Loading

After preprocessing, the cleaned datasets were loaded into a structured SQLite database. The database schema was designed to support efficient SQL queries for fund performance, investor analytics, transaction analysis, and dashboard reporting. This centralized database served as the primary source for EDA, performance analytics, advanced analytics, and the Power BI dashboard.

Figure 2: Raw Dataset Files in data/raw Directory

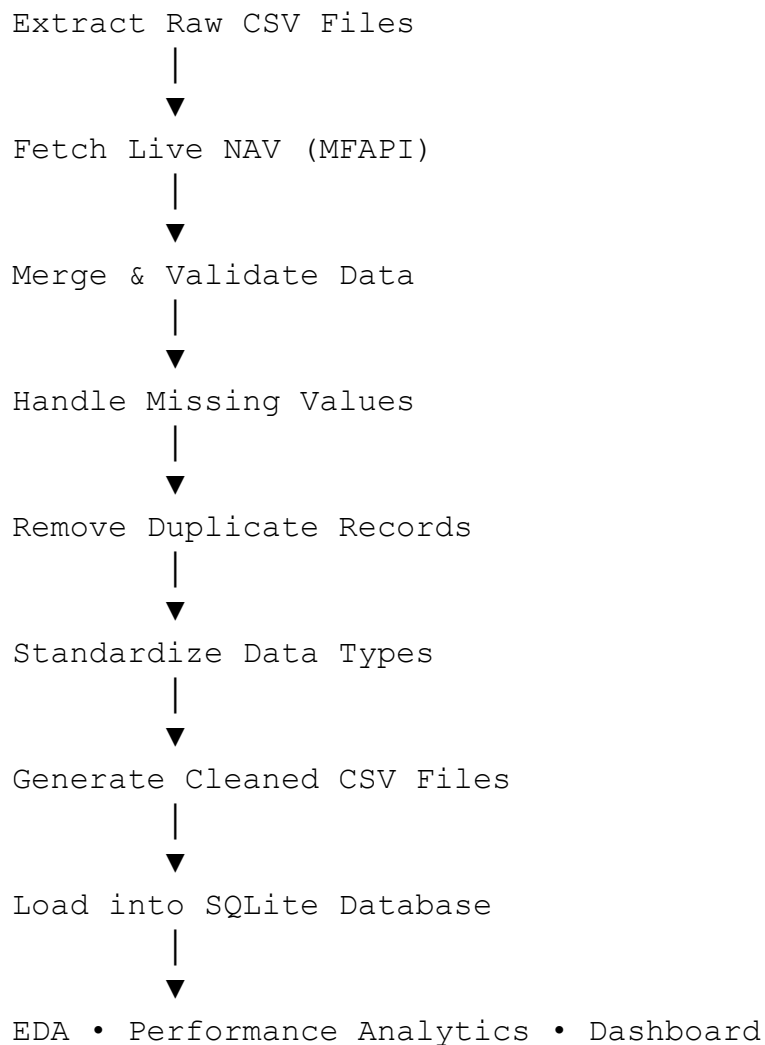
Name	Status	Date modified	Type	Size
 01_fund_master		21-06-2026 23:33	Microsoft Excel Co...	7 KB
 02_nav_history		21-06-2026 23:33	Microsoft Excel Co...	1,191 KB
 03_aum_by_fund_house		21-06-2026 23:33	Microsoft Excel Co...	4 KB
 04_monthly_sip_inflows		21-06-2026 23:33	Microsoft Excel Co...	2 KB
 05_category_inflows		21-06-2026 23:33	Microsoft Excel Co...	4 KB
 06_industry_folio_count		21-06-2026 23:33	Microsoft Excel Co...	1 KB
 07_scheme_performance		21-06-2026 23:33	Microsoft Excel Co...	7 KB
 08_investor_transactions		21-06-2026 23:33	Microsoft Excel Co...	3,055 KB
 09_portfolio_holdings		21-06-2026 23:33	Microsoft Excel Co...	24 KB
 10_benchmark_indices		21-06-2026 23:33	Microsoft Excel Co...	246 KB
 axis_bluechip_nav		02-07-2026 01:38	Microsoft Excel Co...	79 KB
 hdfc_top100_live_nav		22-06-2026 01:05	Microsoft Excel Co...	65 KB
 icici_bluechip_nav		02-07-2026 01:38	Microsoft Excel Co...	69 KB
 kotak_bluechip_nav		02-07-2026 01:38	Microsoft Excel Co...	70 KB
 nippon_largecap_nav		02-07-2026 01:38	Microsoft Excel Co...	69 KB
 sbi_bluechip_nav		02-07-2026 01:38	Microsoft Excel Co...	70 KB

ETL Pipeline & Data Cleaning Process

ETL Pipeline

The Mutual Fund Analytics project implements an Extract, Transform and Load (ETL) pipeline to automate data preparation. Raw datasets and live NAV data are extracted, validated, cleaned and transformed before being loaded into an SQLite database for further analysis.

ETL Workflow



Data Cleaning Process

- Handled missing values and validated records.
- Removed duplicate observations.
- Converted date columns into datetime format.
- Validated NAV, AUM, return and transaction values.
- Standardized categorical values.
- Saved processed datasets into the data/processed folder.

ETL Output

The ETL pipeline generated cleaned datasets, populated the SQLite database, and produced processed files used in SQL analytics, EDA, Performance Analytics, Advanced Analytics and the Power BI dashboard.

Figure 3: ETL Pipeline Scripts

Name	Status	Date modified	Type	Size
 compute_metrics	✓	24-06-2026 15:34	Python Source File	1 KB
 data_cleaning	✓	24-06-2026 15:18	Python Source File	2 KB
 etl_pipeline	✓	22-06-2026 01:16	Python Source File	2 KB
 live_nav_fetch	✓	22-06-2026 01:12	Python Source File	1 KB
 recommender	✓	01-07-2026 23:55	Python Source File	1 KB

SQLite Database Design & SQL Analytics

Database Design

The cleaned datasets were stored in an SQLite relational database to enable efficient storage, retrieval, and analytical querying. Separate tables were created for NAV history, scheme performance, investor transactions, benchmark indices, portfolio holdings, and other supporting datasets. Primary keys and common identifiers such as AMFI Code were used to maintain relationships between datasets.

Database Schema

The database schema was designed to reduce redundancy while supporting fast analytical queries. SQLite was selected because it is lightweight, portable, and well suited for data analytics projects.

SQL Analytics

SQL queries were executed to answer important business questions related to mutual fund performance, investor behaviour, and market trends.

- Top performing mutual funds based on annual return.
- Fund houses with the highest Assets Under Management (AUM).
- Category-wise investment inflows.
- Monthly SIP collection trend.
- Investor transaction distribution by transaction type.
- Top portfolio sectors by allocation.

- Benchmark performance comparison.
- Average return by mutual fund category.
- Highest rated schemes based on Morningstar ratings.
- Risk analysis using Alpha, Beta and Sharpe Ratio.

Key Outcome

The SQL layer provided the foundation for Exploratory Data Analysis, dashboard development, performance analytics, and business reporting by transforming raw data into actionable insights.

Figure 4: SQL Schema and Query Files

Name	Status	Date modified	Type	Size
 queries		24-06-2026 15:44	SQL Text File	2 KB
 schema		24-06-2026 15:40	SQL Text File	1 KB

Performance Analytics

Introduction

Performance Analytics was conducted to evaluate the financial performance of mutual funds using historical return and risk measures. Various quantitative metrics were analysed to identify high-performing funds, compare fund categories, and understand the relationship between returns and investment risk.

The analysis helps investors assess fund quality based on risk-adjusted performance rather than returns alone.

Performance Metrics

Metric	Description
Annual Return	Measures yearly percentage return generated by a fund.
Alpha	Measures excess return compared to benchmark performance.
Beta	Indicates volatility relative to the overall market.
Sharpe Ratio	Measures risk-adjusted return considering total volatility.
Sortino Ratio	Measures downside risk-adjusted return.
Standard Deviation	Represents overall investment risk and volatility.

Maximum Drawdown	Measures the largest decline from peak portfolio value.
Expense Ratio	Annual management fee charged by the fund.

Performance Analysis Process

The historical mutual fund performance dataset was processed to calculate and compare multiple performance indicators. Risk-adjusted metrics were analysed to determine which funds delivered better returns for the level of risk undertaken.

Comparisons were performed across different fund houses, investment categories, and benchmark indices. Statistical summaries and visualizations were generated to highlight performance differences and identify consistently performing mutual funds.

Key Findings

- Large Cap and Flexi Cap funds generated relatively stable long-term returns.
- Higher Sharpe Ratio values indicated superior risk-adjusted performance.
- Funds with lower Maximum Drawdown demonstrated better downside protection.
- Alpha values identified funds outperforming their benchmark indices.
- Lower Expense Ratio funds generally provided improved long-term investment efficiency.
- Risk levels varied significantly across different mutual fund categories.

Figure 5: Normalized Performance Comparison



Conclusion:

The normalized performance comparison shows that different mutual funds exhibited varying growth trajectories over the analysis period. While some funds consistently

outperformed the benchmark indices, others demonstrated relatively stable returns with lower volatility. The comparison highlights the importance of long-term performance evaluation and diversification in mutual fund investment decisions.

Advanced Analytics & Risk Analysis

Introduction

Advanced analytics techniques were implemented to evaluate mutual fund risk, portfolio stability, and investor behaviour. Risk-adjusted metrics and portfolio analysis were used to support data-driven investment decisions.

Advanced Analytics Performed

- Value at Risk (VaR) estimation
- Conditional Value at Risk (CVaR)
- Sharpe Ratio analysis
- Beta and Alpha evaluation
- Maximum Drawdown analysis
- Investor Cohort Analysis
- SIP Continuity Analysis
- Mutual Fund Recommendation Engine

Key Findings

The analysis identified funds with superior risk-adjusted returns and lower downside risk. VaR and CVaR quantified potential losses, while Sharpe Ratio and Maximum Drawdown helped compare fund quality. The recommendation engine suggested suitable funds based on return, risk and consistency metrics.

Figure 6: Rolling 90-Day Sharpe Ratio



Conclusion:

The rolling 90-day Sharpe Ratio indicates changes in risk-adjusted performance over time. Higher Sharpe Ratio values represent better returns for the level of risk undertaken, whereas lower or negative values indicate weaker risk-adjusted performance. This analysis helps identify funds that consistently delivered superior returns while maintaining controlled risk.

Interactive Power BI Dashboard

Dashboard Overview

An interactive four-page Power BI dashboard was developed to present key insights derived from the Mutual Fund Analytics project. The dashboard enables users to explore fund performance, investor behaviour, market trends, and portfolio statistics through dynamic visualisations and slicers.

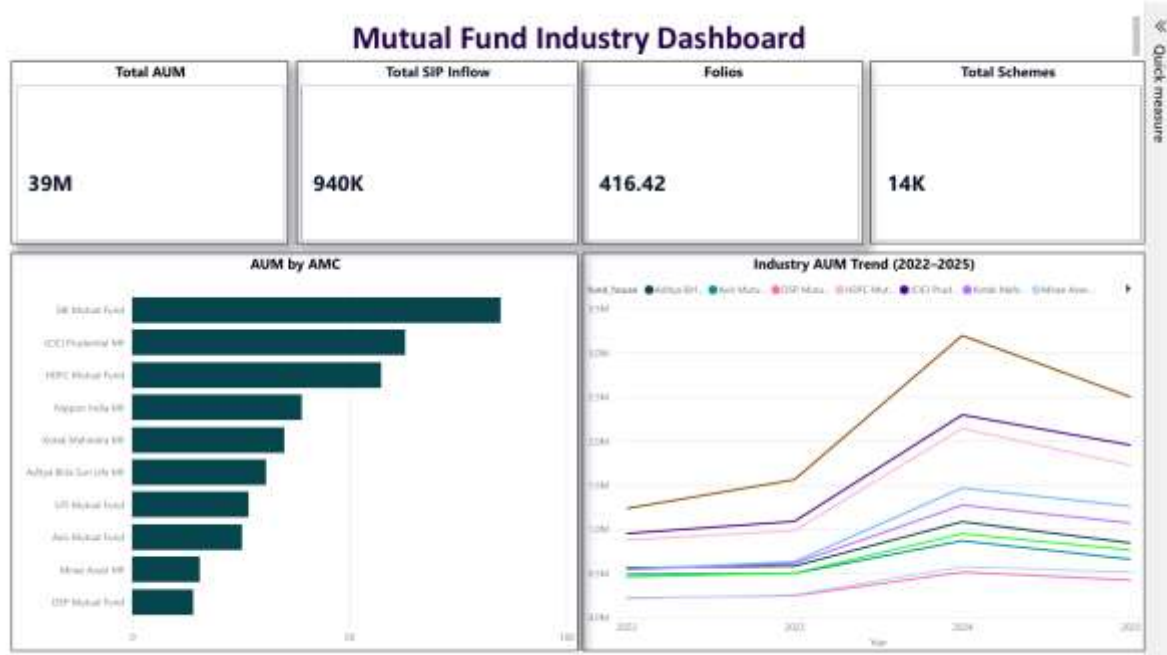
Dashboard Pages

Dashboard Page	Description
Industry Overview	KPIs including Total AUM, SIP Inflows, Folios, Schemes, AUM trends and AMC comparison.
Fund Performance	Normalized fund performance, benchmark comparison, risk vs return analysis and fund scorecard.
Investor Analytics	Transaction distribution, age-group analysis, state-wise investment and monthly transaction trends.
SIP & Market Trends	SIP inflows versus market indices, category inflows and market trend analysis.

Key Dashboard Features

- Interactive slicers for Fund House, Category and Time Period
- Drill-down visualizations
- KPI cards for major financial indicators
- Benchmark comparison charts
- Professional business dashboard layout

Figure 7 – Mutual Fund Industry Overview Dashboard



Conclusion

The dashboard transforms complex mutual fund data into an intuitive decision-support system. Interactive filters and visualisations enable investors and analysts to quickly identify trends, compare schemes, and evaluate investment performance.

Business Insights & Recommendations

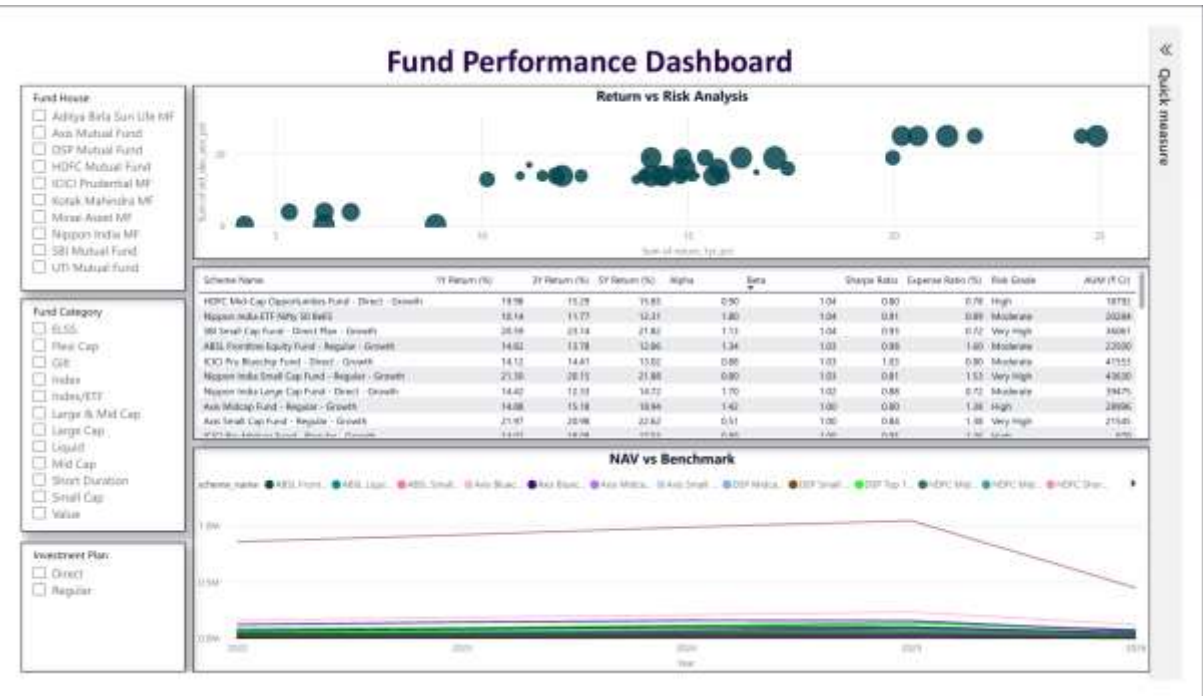
Key Business Insights

- Equity-oriented mutual funds delivered comparatively stronger long-term returns than most debt-oriented categories.
- SIP inflows increased consistently, reflecting growing investor participation and long-term investment confidence.
- Large Cap and Flexi Cap funds demonstrated relatively stable performance with controlled volatility.
- Risk-adjusted metrics such as Sharpe Ratio and Alpha helped identify consistently performing schemes.
- Benchmark comparison highlighted several funds that outperformed market indices over the analysis period.

Business Recommendations

- Diversify investments across multiple fund categories to reduce portfolio risk.
- Prefer funds with higher Sharpe Ratio and consistent Alpha for long-term investing.
- Adopt SIP investments to benefit from rupee-cost averaging and disciplined investing.
- Review portfolio allocation periodically based on changing market conditions.
- Compare fund performance with benchmark indices before making investment decisions.

Figure 8: Fund Performance Dashboard



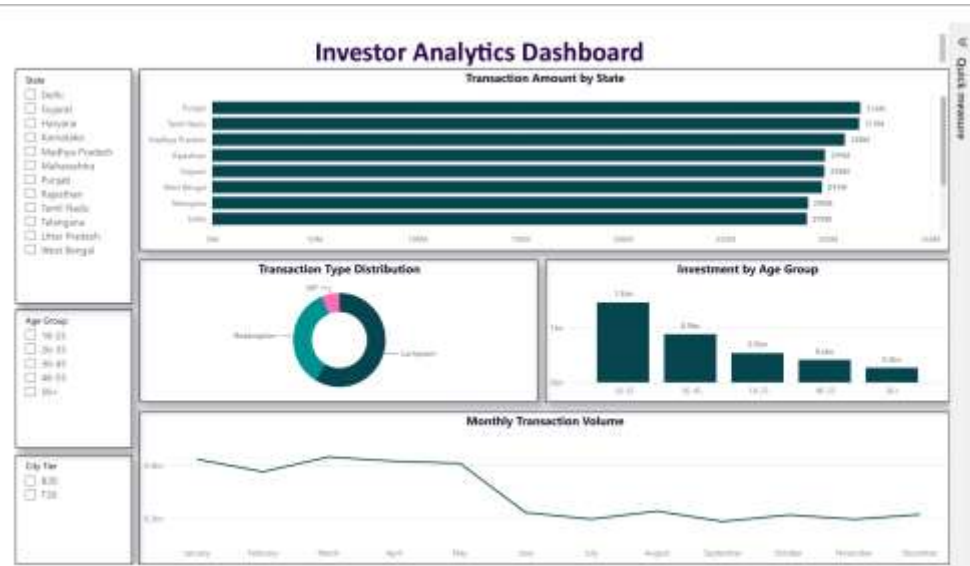
Conclusion

The business insights generated from this project help investors understand fund performance, risk characteristics, and investment behaviour. These recommendations support better portfolio management and informed financial decision-making.

Investor Analytics Dashboard

The Investor Analytics dashboard focuses on analysing investor behaviour across different states, age groups, transaction types, and monthly investment trends. It enables detailed demographic analysis and investment pattern identification.

Figure 9 – Investor Analytics Dashboard



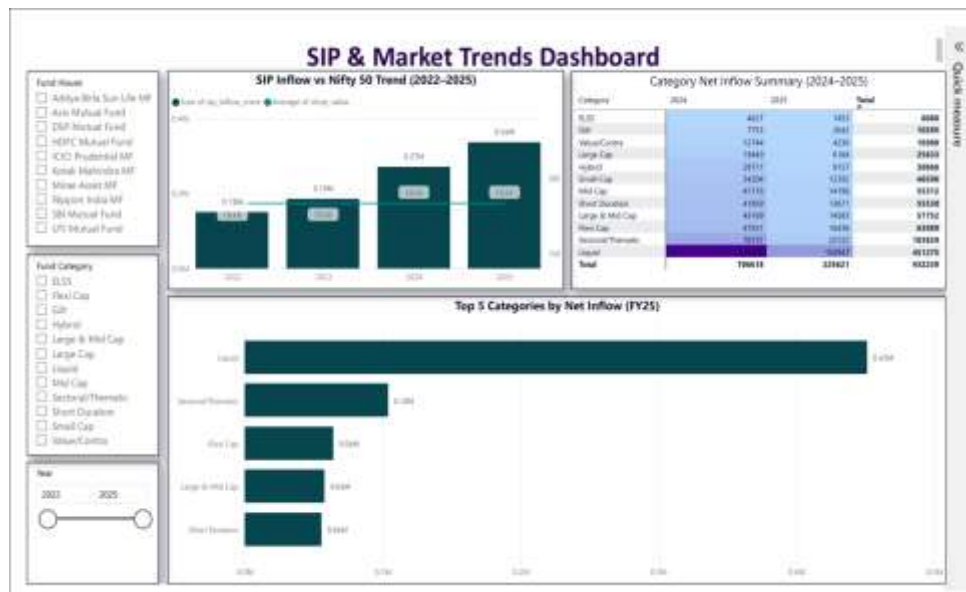
Conclusion

The Investor Analytics dashboard highlights investor demographics and transaction behaviour. It helps identify investment preferences across age groups, states, and transaction types, enabling better customer segmentation.

SIP & Market Trends Dashboard

The SIP & Market Trends dashboard compares SIP inflows with benchmark market indices and category-wise investment trends. It enables analysis of long-term investment behaviour and market performance.

Figure 10 – SIP & Market Trends Dashboard



Conclusion

The dashboard illustrates the relationship between SIP inflows and market movements while highlighting category-wise investment trends. These insights support long-term investment planning and market analysis.

References

This project was developed using publicly available datasets, official documentation, and widely adopted data science tools.

- MFAPI (<https://www.mfapi.in/>) – Live Mutual Fund NAV Data
- AMFI India – Mutual Fund Industry Information
- Microsoft Power BI Documentation
- SQLite Documentation
- Pandas Documentation
- NumPy Documentation
- Matplotlib Documentation
- Python Documentation
- Scikit-learn Documentation
- Git & GitHub Documentation

Project Deliverables Submitted

- ETL Pipeline Scripts
- SQLite Database and SQL Queries
- EDA Notebook
- Performance Analytics Notebook
- Advanced Analytics Notebook
- Power BI Dashboard (.pbix)
- Final Project Report
- Presentation Slides
- GitHub Repository

Acknowledgement

I sincerely thank Bluestock Fintech for providing me the opportunity to work on the Mutual Fund Analytics Capstone Project. This project significantly enhanced my practical knowledge of Python, SQL, SQLite, ETL pipelines, data analytics, financial analytics, Power BI dashboard development, and business intelligence. I am grateful for the guidance, resources, and learning experience that helped me successfully complete this project.

End of Report

This report presents the complete end-to-end implementation of the Mutual Fund Analytics Capstone Project. All objectives, analyses, dashboards, and business insights have been successfully completed.